

Open Charge Point Protocol (OCPP)

Key - Words

OCPP, EV Charging Infrastructure, Interoperability, Smart Charging, Load Management, ISO 15118, OCPP 2.0.1.

1. Definition and Purpose of OCPP

The Open Charge Point Protocol (OCPP) is an open-source communication protocol designed to ensure interoperability between electric vehicle (EV) charging stations (Charge Points) and central management systems (Central Systems or CSMS). It enables seamless communication between hardware from different manufacturers and backend software platform.

OCPP allows charge point operators to remotely monitor and manage charging stations, authorize users, start and stop charging sessions, update firmware, configure devices, manage billing systems, and implement smart charging functionalities. Through standardization, OCPP supports scalability, flexibility, and efficient operation of EV charging networks.

2. Historical Development of OCPP

OCPP was developed in 2009 by the Dutch company E-laad in collaboration with industry partners. The first official release, OCPP 1.0, appeared in 2010, focusing on basic communication between charging stations and backend systems.

In 2013, the Open Charge Alliance (OCA) was established to promote global adoption and standardization of OCPP.

The protocol evolved through multiple versions:

OCPP 1.2 (2012)

OCPP 1.5 (2015)

OCPP 1.6 (widely adopted version)

OCPP 2.0 (2018)

OCPP 2.0.1 (2019/2020)

OCPP 2.0.1 introduced significant improvements including enhanced security, improved transaction handling, device management capabilities, extended smart charging functions, and native support for ISO 15118. Unlike earlier versions, OCPP 2.0 is not backward compatible with OCPP 1.6.

Today, **OCPP is recognized as the industry standard for EV charging communication.**

3. Architecture and Communication Model

OCPP defines two primary roles:

Charge Point (Client): The physical EV charging station.

Central System (Server/CSMS): The backend management platform.

Communication occurs through standardized message exchanges.

Two implementation methods exist:

SOAP (XML-based communication)

JSON over WebSocket (lighter and more efficient; currently preferred)

OCPP 2.0.1 enhances security by supporting encrypted communication (TLS), secure firmware updates, certificate management, and authentication mechanisms.

4. Functional Blocks of OCPP 2.0.1

OCPP 2.0.1 is structured into functional blocks that define protocol behavior:

Security	Transactions	Tariff and Cost	ISO 15118 Certificate Management
Provisioning	Remote Control	Metering	Diagnostics
Authorization	Availability	Smart Charging	Display Message
Local Authorization List Management	Reservation	Firmware Management	Data Transfer

Each **functional block** defines standardized use cases, message formats, configuration keys, and testing procedures to ensure reliable implementation.

5. Integration with Other Protocols

OCPP can interface with complementary standards to enhance functionality:

ISO 15118: Enables Plug & Charge and Vehicle-to-Grid (V2G) applications.

OpenADR: Enables automated demand response communication with utilities.

This integration improves interoperability, grid stability, and energy optimization.

6. Key Benefits of OCPP

The main advantages of OCPP include:

Interoperability between multi-vendor hardware and software

Remote monitoring and management

Improved customer experience (authentication, billing, mobile integration)

Operational cost reduction

Dynamic pricing and tariff optimization

Future-proof infrastructure

Enhanced cybersecurity (OCPP 2.0.1)

7. Main Use Cases

OCPP supports several critical operational scenarios:

Remote start/stop of charging sessions

Real-time status and meter value reporting

Load balancing and smart charging control

Firmware updates and diagnostics

Regulatory compliance (e.g., UK EV Smart Charging Regulations, German Eichrecht)

8. Implementation Considerations

OCPP can be implemented either through custom development or by adopting ready-made EV charging management platforms. Due to the complexity of EV infrastructure and regulatory requirements, ready-made solutions are often preferred.

Version compatibility must be carefully managed, as newer OCPP versions are not backward compatible with older ones.

Maintenance and upgrades are essential to ensure long-term efficiency and compliance.

Conclusion

OCPP has evolved into the global standard communication protocol for EV charging infrastructure. Its development reflects the increasing need for interoperability, cybersecurity, smart charging capabilities, and grid integration. With OCPP 2.0.1, the protocol provides enhanced security, improved transaction management, advanced smart charging functions, and native ISO 15118 integration, positioning it as a future-ready framework for scalable and sustainable EV charging networks.

Reference:

<https://www.ampeco.com/guides/complete-ocpp-guide/>